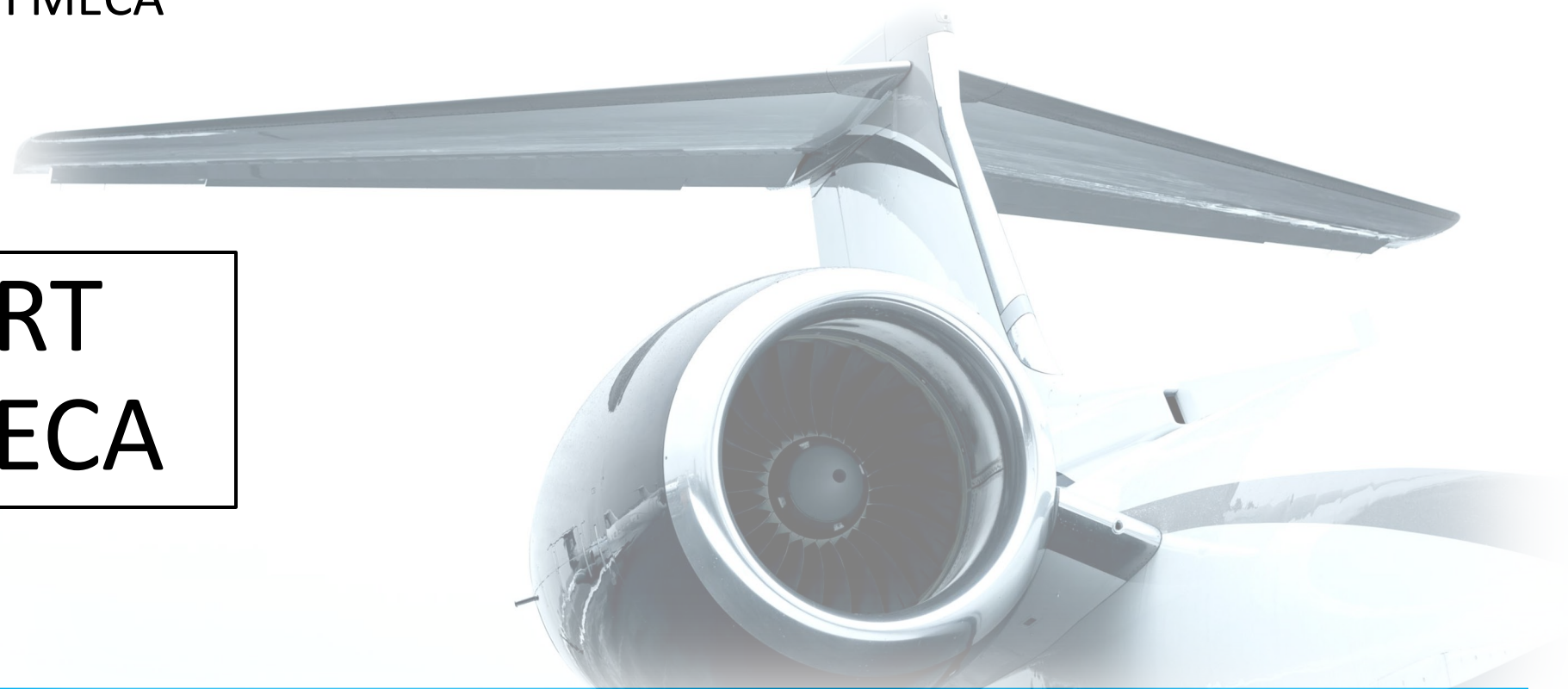


Safety and Reliability Training

Module 2: RELIABILITY ENGINEERING

Submodule 2.B: FMEA/FMECA

PIECE PART
FMEA/FMECA





Objectives of this training

- ✓ To learn the methodology of Piece Part FMEA





Outline

- ☐ FMEA Introduction
- ☐ PPFMEA – Introduction
- ☐ PPFMEA – Ground Rules and Assumptions
- ☐ PPFMEA – Model
- ☐ PPFMEA – Block Building
- ☐ PPFMEA – Block Functions
- ☐ PPFMEA – Block Reliability Model
- ☐ PPFMEA – Parts and COTS





Outline

- ❑ PPFMEA – Part/COTS Failure Mode Effects
- ❑ PPFMEA – Part/COTS Failure Mode Failure Rate
- ❑ PPFMEA – Part/COTS Failure Mode Causes
- ❑ PPFMEA – Failure Mode Detection
- ❑ PPFMEA – Criticality Analysis
- ❑ PPFMEA – Analysis Taxonomy
- ❑ PPFMEA – Tools
- ❑ PPFMEA Examples



PPFMEA – Introduction (I)

- What a Piece-Part FMEA is?? Is it dangerous?



PPFMEA – Introduction (II)

- The Failure Mode and Effect Analysis (FMEA) is a powerful dependability technique to evaluate systems and equipment inherent potentiality to fail leading to effects.
- FMEA methodology is based on the identification of the potential failure modes of each item according to analyst knowledge and engineering judgement, and their effects on the upper system levels.
- According to the analysis depth, FMEAs can be classified in:
 - Functional (FFMEA) – down to functional block level
 - Piece-Part (PPFMEA) – down to commercial hardware level



PPFMEA – Introduction (III)

- A PPFMEA analyses system or equipment indenture levels down to commercial hardware components (parts or COTS).
- As opposite to FFMEA, blocks, parts and COTS have determined, although not necessarily available, failure data.
- A PPFMEA is adequate:
 - When equipment or system design is mature.
 - When the availability of time and/or resources for the analysis are sufficient.
 - As a continuation analysis of a previous FFMEA.



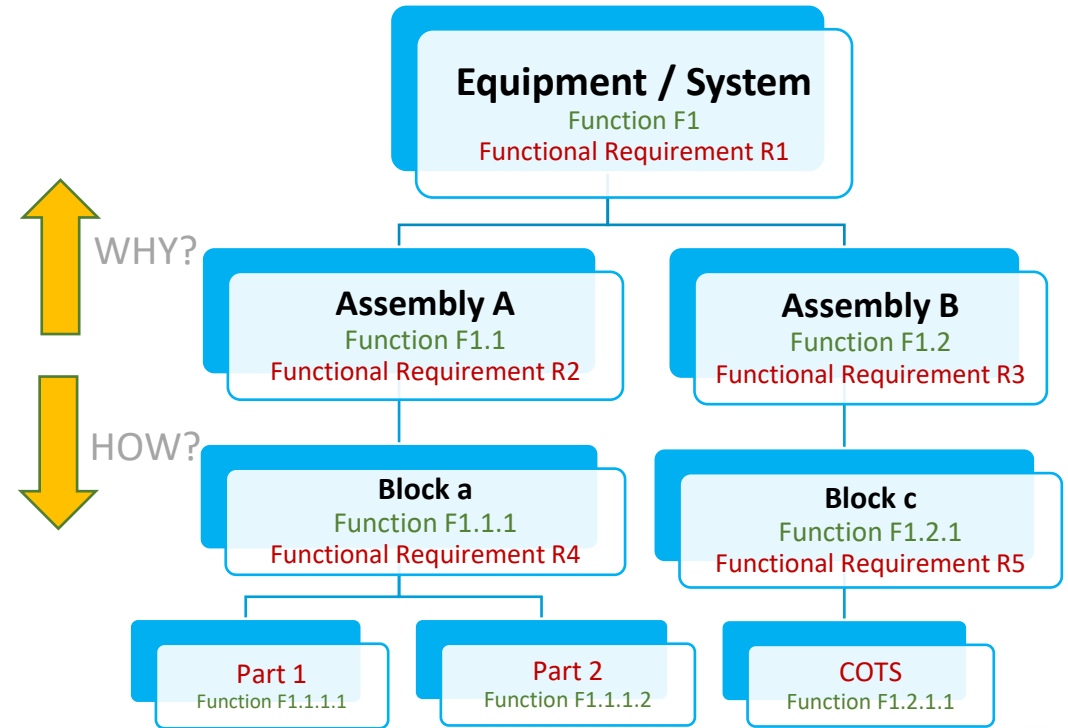
PPFMEA – Grounds rules and Assumptions

- Similarly to other FMEA types, the analysis must be based on ground rules that determine its essential aspects, such as:
 - ✓ Team and responsibilities.
 - ✓ Purpose, according to client necessities.
 - ✓ Scope, including boundaries and interfaces.
 - ✓ Failure and success criteria.
 - ✓ Tool (spreadsheet template or WRR).
- Some assumptions related to the current project are normally required to tailor the analysis. Some examples are:
 - ✓ Taxonomy (it will be further explained later in this presentation).
 - ✓ Rating scales, if necessary (e.g. for severity, frequency...etc.).



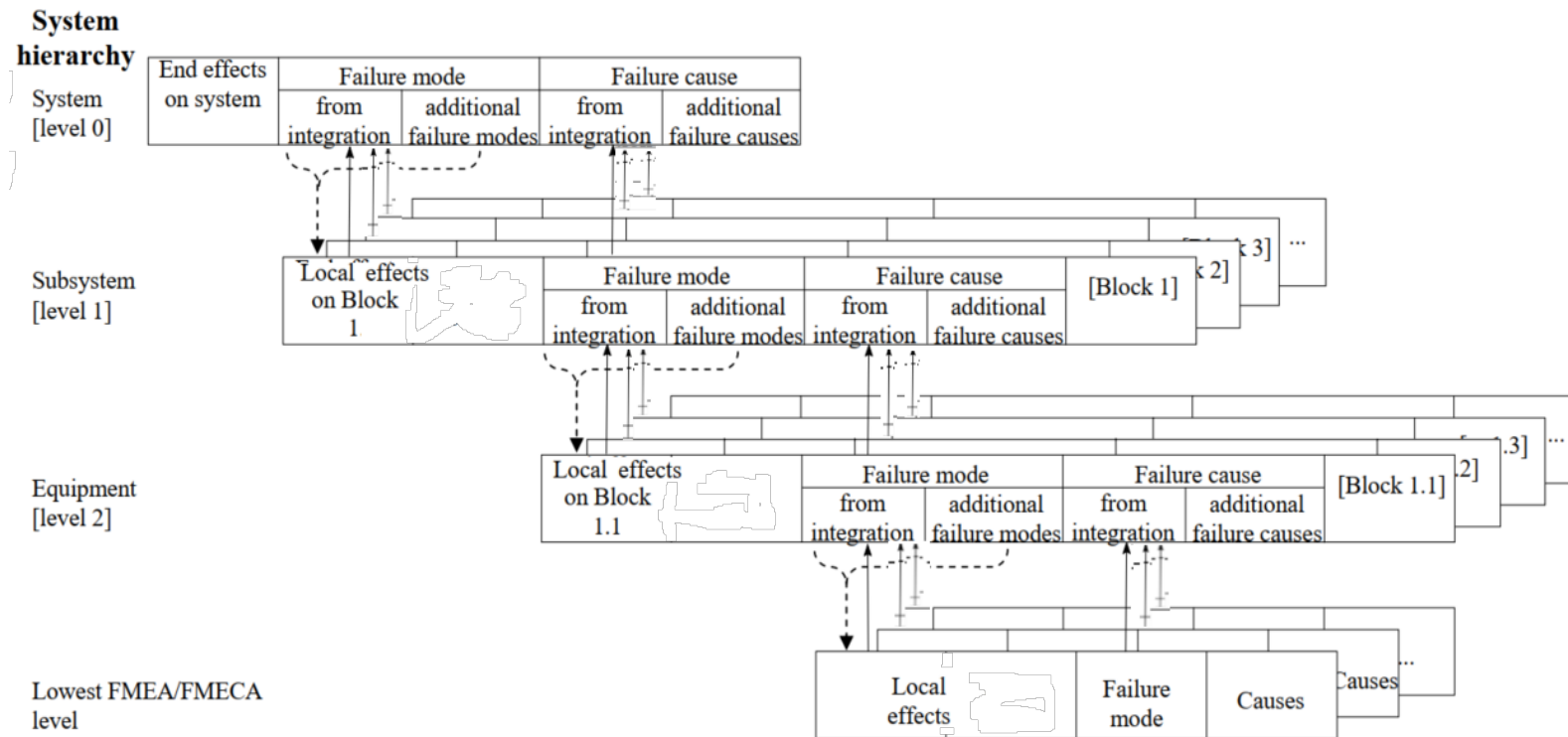
PPFMEA – Model (1)

- The PPFMEA model, similarly to FFMEA, is initially based on a hierarchic breakdown of the equipment / system functions as conceived according to its corresponding requirements. However, parts and COTS are finally included.
- Functions and requirements can be allocated to intermediate functional failures (blocks). At part level only functions are relevant, usually.



PPFMEA – Model (2)

- Similarly to other FMEA types, causes, failure modes and effects must keep their mutual relationships along the analysis.



PPFMEA – Model (3)

- Several indenture levels are identified, each of them composed of hardware item (assemblies, parts, COTS) or functional blocks.
- Definition of levels and functional blocks (if any) should consider that:
 - ✓ A reduced quantity of them does not lead to an easier analysis since it would result in complex blocks, difficult to deal with.
 - ✓ On the other hand, a high number would make the analysis quite extensive and complex.
- Therefore, a trade off between quantity and complexity must be reached for the sake of the analysis.



PPFMEA – Block building

- Functional Blocks, if used, are intermediate PPFMEA elements created for better equipment/system modelling. They can correspond to those created for previous FFMEA.
- Blocks can perform one or several functions.
- As a general rule for electronic equipment, a block is built from an integrated circuit, if any, and its related components. However, others such as those related to power decoupling or equipment interfaces are built separately because of their criticality.
- It is advisable that blocks in the same indenture level are independent (not sharing parts/COTS).



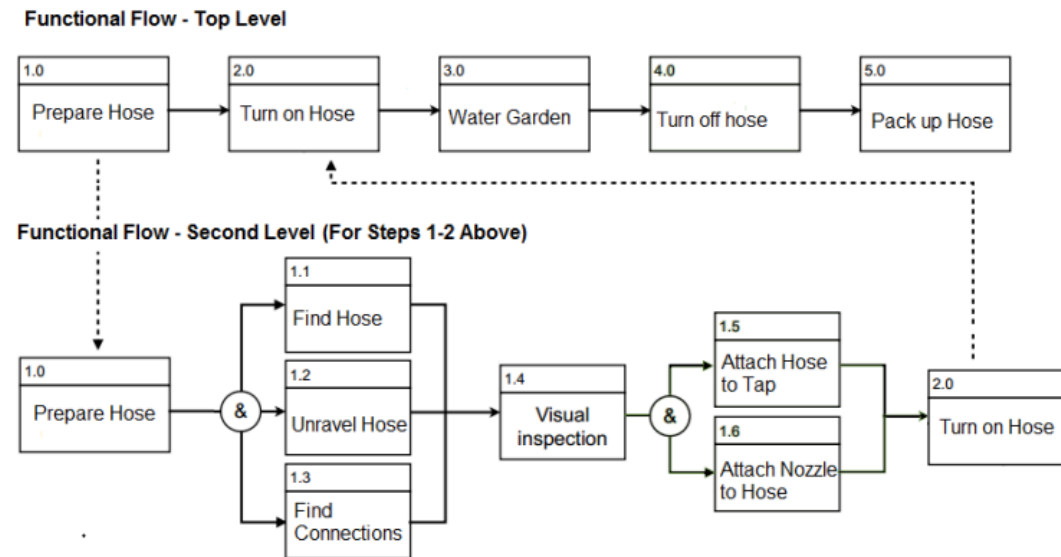
PPFMEA – Block Functions (1)

- Block functions can be established according to:
 - Allocated functions of the immediate upper-level block.
 - Input and output interfaces:
 - ✓ Electrical signals such as power supplies, grounds, analog / digital inputs or outputs...etc.,
 - ✓ Data connections like I2C, SPI...etc.
 - ✓ Physical interchange such as heating, cooling...etc.
 - ✓ Human interactions like visual, auditory...etc.
 - Presence of key components (microcircuits) inside which feature the block functionality.



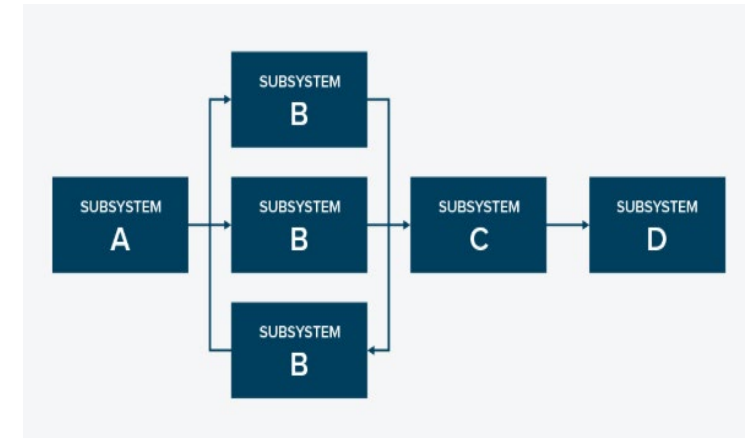
PPFMEA – Block Functions (2)

- Block functions provide flexibility to the model since they act as an intermediate level between block and its parts by establishing functional relationships among these ones.
- Block functions must be hierarchically coherent with those of the other blocks at upper, lower and same indenture level in the functional model.



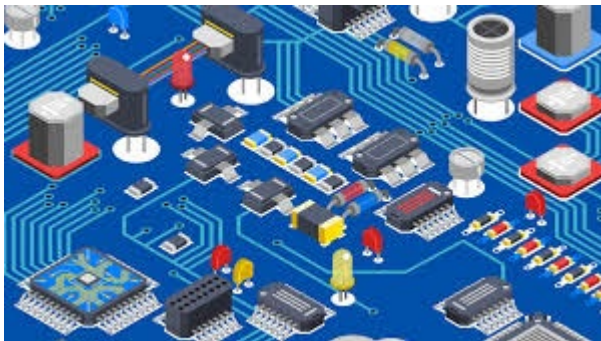
PPFMEA – Block Reliability model

- It is very advisable to analyse the reliability relationships between blocks. This is to say, identify the functional path(s) across blocks which each high-level function requires to be accomplished.
- This task highlights:
 - Single point failures which may impede equipment functions
 - Functional redundancies (fail-tolerant architectures).
- Each functionally redundant channel should be modelled as a higher-level block composed by one or several lower-level ones.



PPFMEA – Parts and COTS

- Parts and COTS are the simplest PPFMEA elements.
- According to their functionality, they can be:
 - Simple, carrying out just one function by means of a reduced set of or interfaces (e.g. resistors, capacitors, mechanical components,...etc.)
 - Complex, performing several functions on several interfaces (e.g. microcircuits, connectors, COTS...etc.).



PPFMEA – Part/COTS Failure Modes (1)

- Part/COTS, as defined elements, have determined failure modes. However, their determination is quite different depending on component functional complexity.
- As a general rule, their failure modes can be obtained from:
 - Manufacturer (rarely).
 - Field data (even more rarely).
 - Analysis (for complex components). A special case are “Pin FMEAs”.
 - Standards such as MIL-HDBK-338B or ECSS-Q-ST-30-02C.
 - Commercial libraries such as FMD.
 - Engineering knowledge.
 - Assumptions.



PPFMEA – Part/COTS Failure Modes (2)

- Simple components have just one function and a few failure modes for each type which can be easily obtained, estimated or extracted.
- Complex components such as microcircuits, COTS...etc., can have several functions and therefore a considerable quantity of failure modes. They can be analysed as functional blocks similarly to FMEA.
- An adequate taxonomy of part/COTS failure modes is highly advisable to keep analysis traceability and modularity.



PPFMEA – Part/COTS Failure Mode Effects (1)

- Functional effects of each part/COTS failure mode are evaluated at several indenture levels:
 - LOCAL – related to the effects led in the output of the part/COTS (WRR© considers, specifically, the failure mode of the immediate upper-level one).
 - NEXT (optional) – related to the failure modes led in an intermediate upper item (functional block or physical item).
 - END - related to the failure modes of the highest level under analysis (equipment or system).



PPFMEA – Part/COTS Failure Mode Effects (2)

- Special care must be taken to propagate coherently the failures effects upwards, rather than forwards, according to the functional model.
- However, in some case, (e.g. shorted input), there should be also taken into account the consequences on previous blocks when analysing. For example, these could avoid partially or totally powering up of other equipment area.
- An adequate taxonomy of failure effects is highly advisable to keep both analysis traceability and modularity.



PPFMEA – Part/COTS Failure Mode Effects(3)

- Normally, each Failure Mode is linked to an effect. However, different effects can be assumed to occur in certain cases such as:
 - For avoiding an excessive quantity of part/COTS failure modes (e.g. when managing a great quantity of signals).
 - When effects depend on:
 - Equipment phases.
 - Equipment states (operating, non-operating,...etc.).
 - External events (transients, inrush currents, human behaviour,...etc.).
- In these cases, a conditional probability usually expressed as a percentage can be included in the effect FR computation:

$$\text{EffectFR} = \text{ModeFR} * P(E/M)$$



PPFMEA – Part/COTS Failure Mode Failure Rate (1)

- For simple components, FM Failure rate is obtained by applying the corresponding “Mode ratio” to the total component failure rate.

$$\text{ModeFR} = \text{Part/COTS FR} * \text{Mode Ratio}$$

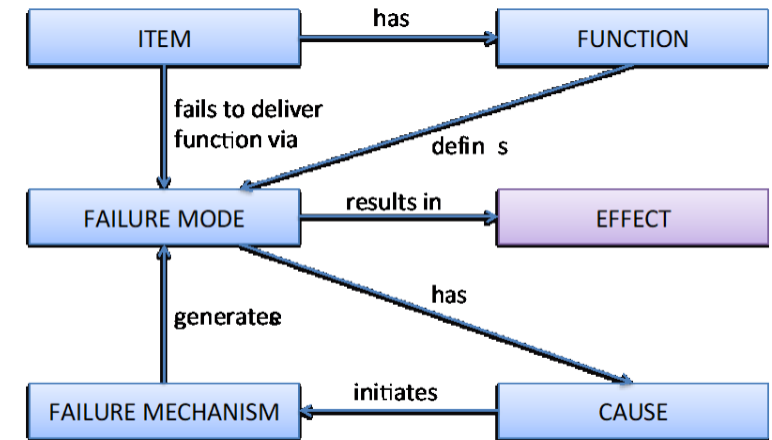
- In case of complex components, if FM distribution is not available, FM Failure rate can be calculated by using some assumptions and the following expression:
 - Part/COTS functions are proportionally weighted about it.
 - Failure Modes are equally distributed for each function.

$$\text{ModeFR} = \text{Part/COTS FR} * \text{Function Ratio} * \text{Mode Ratio}$$



PPFMEA – Part/COTS Failure Mode Causes

- Similarly to other FMEAs, immediate lower-level failure modes contribute to under-analysis failure mode occurrence as causes.
- Failure mode causes lead to failure mechanisms, such as “fatigue” or “corrosion” which in turn results in failure modes.
- Whether included or not in the PPFMEA, causes identification and mitigation allows for control of failure mode frequency.
- At part/COTS level, causes and mechanisms are normally not considered.



PPFMEA – Failure Mode Detection

- As far as possible, parts/COTS involved in failure detection functions should be assigned to specific blocks in the functional model.
- These functions are not normally included in the main equipment ones. However, their functional failures can be relevant since:
 - ✓ Main Function Failures Modes can get undetected (latent).
 - ✓ Significant effects can be caused if related main function also fails. Multiple failures are, at first, out of the PPFMEA scope. However, for the sake of following analyses, such as FTA, their potentiality should be clearly indicated.
 - ✓ Testability requirements (e.g. Failure Detection Rate) may be imposed.



PPFMEA – Criticality Analysis

- Criticality analysis is a systematic approach to evaluate and prioritize assets based on the severity and likelihood of their potential failures modes.
- It is not common to include a Criticality analysis in the PPFMEA we usually do.
- If required, any of the existing methodologies (RPN, Severity-frequency matrix,...etc.) can be used.

		Consequence Scale				
Probability Index	Consequence Index	1	2	3	4	5
	5	Crit, 13	Crit, 12, 11, 10, 9, 8, 7	Crit, 6, 5, 4	Crit, 3, 2	Crit, 1
Probability Scale	4	Crit, 14, 13	Crit, 12, 11, 10	Crit, 9, 8, 7	Crit, 6, 5, 4	Crit, 3, 2
	3	Crit, 15, 14, 13	Crit, 12	Crit, 12, 11, 10	Crit, 9, 8, 7	Crit, 6, 5, 4
	2	Crit, 17, 16	Crit, 15, 14, 13	Crit, 13	Crit, 12, 11, 10	Crit, 9, 8, 7
	1	Crit, 18	Crit, 17, 16	Crit, 15, 14	Crit, 13	Crit, 12, 11, 10



PPFMEA – Analysis Taxonomy (1)

- An adequate taxonomy of the PPFMEA objects; functions, blocks, failure modes, caused, effects and detection means is highly recommended to keep both analysis traceability and modularity.
- Some of the FMEA taxonomy related techniques can be:
 - ✓ An efficient and scalable coding, considering both client necessities and ANZEN experience. For example:

B01
B01.01
B01.01F01
B01.01F01M01
B01.01F01M01E01
...etc.

WRR tool does not allow two items to have the coding so their scope should be included (system/equipment, assembly, block, part).



PPFMEA – Analysis Taxonomy (2)

- ✓ Functions should be written as a simple noun phrase that defines what action has to be carried out.
- ✓ Failure modes should be expressed according to its function context (e.g. incorporating a word which denotes how it fails) without ambiguity. Requirements provide the function acceptable threshold.
- ✓ Failure effects (and causes, is applicable) are failure modes at higher (or lower) levels so same considerations apply.



PPFMEA –Tools - Spreadsheet

- For non-complex FFMEAs, tailored spreadsheets can be prepared quickly from ANZEN existing templates which are based on previous experience.
- As a general rule, templates provide:
 - ✓ Fields for coding and denomination of analysis objects and other auxiliary for signals, comments and explanations.
 - ✓ Lists for avoiding different spelling
 - ✓ Calculations for quantitative results.



PPFMEA –Tools - Spreadsheet

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	0 empty	0 empty 22 IN/D	0 empty 0 IN/D	0 empty 0 IN/D	0 empty 0 IN/D	0 empty	0 empty	0 empty 0 IN/D	8466 empty 0 IN/D	8466 empty 0 N/A filled Progress: 0.02%	8466 empty 0 IN/D	8466 empty 0 IN/D	8466 empty 0 IN/D	8466 empty 0 IN/D	8466 empty 0 IN/D
	Schematic Reference	Item/Component	Description	Failure Ref.	Component Type	PIN	Failure Mode	Failure Cause	Operational Mode	Failure Effect on Local Component	Failure Effect on Board	Failure Effect on System	Ref - Failure Effect on Local Component	Ref - Failure Effect on Board	Ref - Failure Effect on System
3	C1	Def: 470NPD0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.1.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT	Maintenance	No effect	No effect	No effect	AVIO2353-L3.10.1	AVIO2353-L2.10.1	FLAGS-L3.01.1
4	C1	Def: 470NPD0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.1.2	Ceramic Chip Capacitor	N/A	Shorted	Ceramic Chip Capacitor SHORTED							
5	C1	Def: 470NPD0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.1.3	Ceramic Chip Capacitor	N/A	Opened	Ceramic Chip Capacitor OPENED							
6	C10	Def: 470NPD0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: ARINC, RSxxx Schematic Sheet: 13-FPGA_BANK_33_34_35	10.20.1.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT	Maintenance	No effect (Decoupling Capacitor)	No effect	No effect	AVIO2353-L3.10.4	AVIO2353-L2.10.1	FLAGS-L3.01.1
7	C10	Def: 470NPD0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: ARINC, RSxxx Schematic Sheet: 13-FPGA_BANK_33_34_35	10.20.1.2	Ceramic Chip Capacitor	N/A	Shorted	Ceramic Chip Capacitor SHORTED							
8	C10	Def: 470NPD0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: ARINC, RSxxx Schematic Sheet: 13-FPGA_BANK_33_34_35	10.20.1.3	Ceramic Chip Capacitor	N/A	Opened	Ceramic Chip Capacitor OPENED							
9	C100	Def: 100NPD0402_50V 0402 C1005K7R1H104K R10000 P/N: 011.830-PF	Function: PCI Schematic Sheet: 22-PEX8718_IO	10.30.1.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT							
10	C100	Def: 100NPD0402_50V 0402 C1005K7R1H104K R10000 P/N: 011.830-PF	Function: PCI Schematic Sheet: 22-PEX8718_IO	10.30.1.2	Ceramic Chip Capacitor	N/A	Shorted	Ceramic Chip Capacitor SHORTED							
11	C100	Def: 100NPD0402_50V 0402 C1005K7R1H104K R10000 P/N: 011.830-PF	Function: PCI Schematic Sheet: 22-PEX8718_IO	10.30.1.3	Ceramic Chip Capacitor	N/A	Opened	Ceramic Chip Capacitor OPENED							
12	C101	Def: 100UF1206_6V3 1206 C1206C107M9PACTU R2000 P/N: 011.161-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.2.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT							
13	C101	Def: 100UF1206_6V3 1206 C1206C107M9PACTU R2000 P/N: 011.161-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.2.2	Ceramic Chip Capacitor	N/A	Shorted	Ceramic Chip Capacitor SHORTED							
14	C101	Def: 100UF1206_6V3 1206 C1206C107M9PACTU R2000 P/N: 011.161-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.2.3	Ceramic Chip Capacitor	N/A	Opened	Ceramic Chip Capacitor OPENED							
15	C102	Def: 4.7UF0603_10V 0603 C0603C475M9PACTU R4000 P/N: 011.806-PF	Function: ARINC, RSxxx Schematic Sheet: 14-FPGA_BANK_12_13_14	10.20.2.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT							
		Def: 4.7UF0603_10V 0603 C0603C475M9PACTU	Function: ARINC, RSxxx					Ceramic Chip Capacitor							



PPFMEA –Tools - WRR

- Windchill Risk & Reliability (WRR) is an integrated suite of software tools used for analyzing product performance and reliability. Some of FMEA module main features are:
 - ✓ FFMEA and PPFMEA capabilities.
 - ✓ FMEA building using either industry standards or user-defined ones.
 - ✓ Failure mode libraries for electronic and mechanical parts.
 - ✓ Automatic building (top-down and down-top) of FMEA levels and Fault Trees.
 - ✓ Error verification.
 - ✓ Built-in and user-defined calculations.
 - ✓ Reports and graphs generation according to user necessities.
 - ✓ Data links from Reliability Prediction.



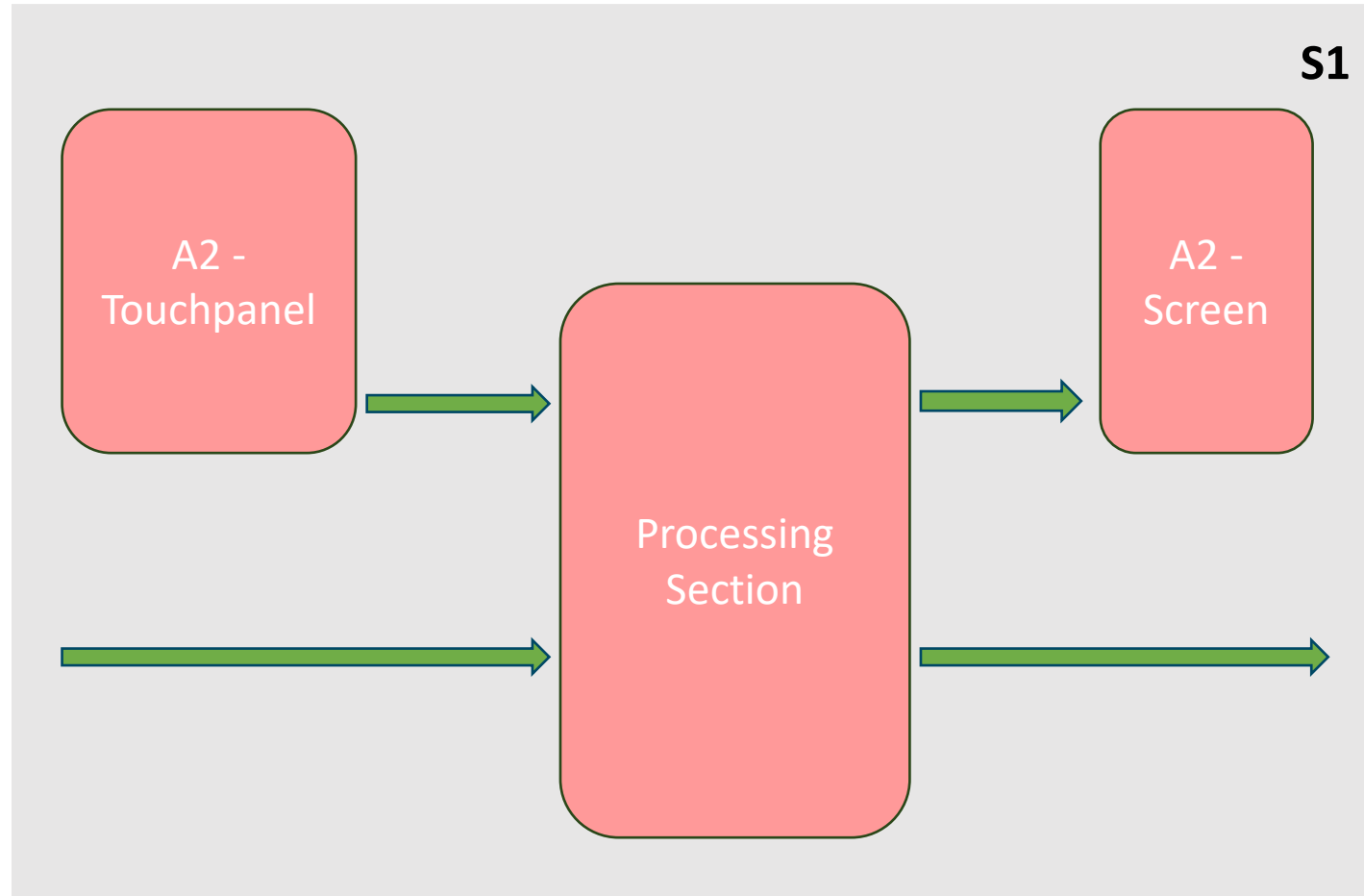
PPFMEA Examples – Introduction

- The example PPFMEA completes a preliminary processing module FFMEA.
- Design is mature to part level.
- Functional model is built from equipment knowledge, identifying four indenture levels:
 - 1º - Equipment
 - 2º - Assemblies
 - 3º - Parts

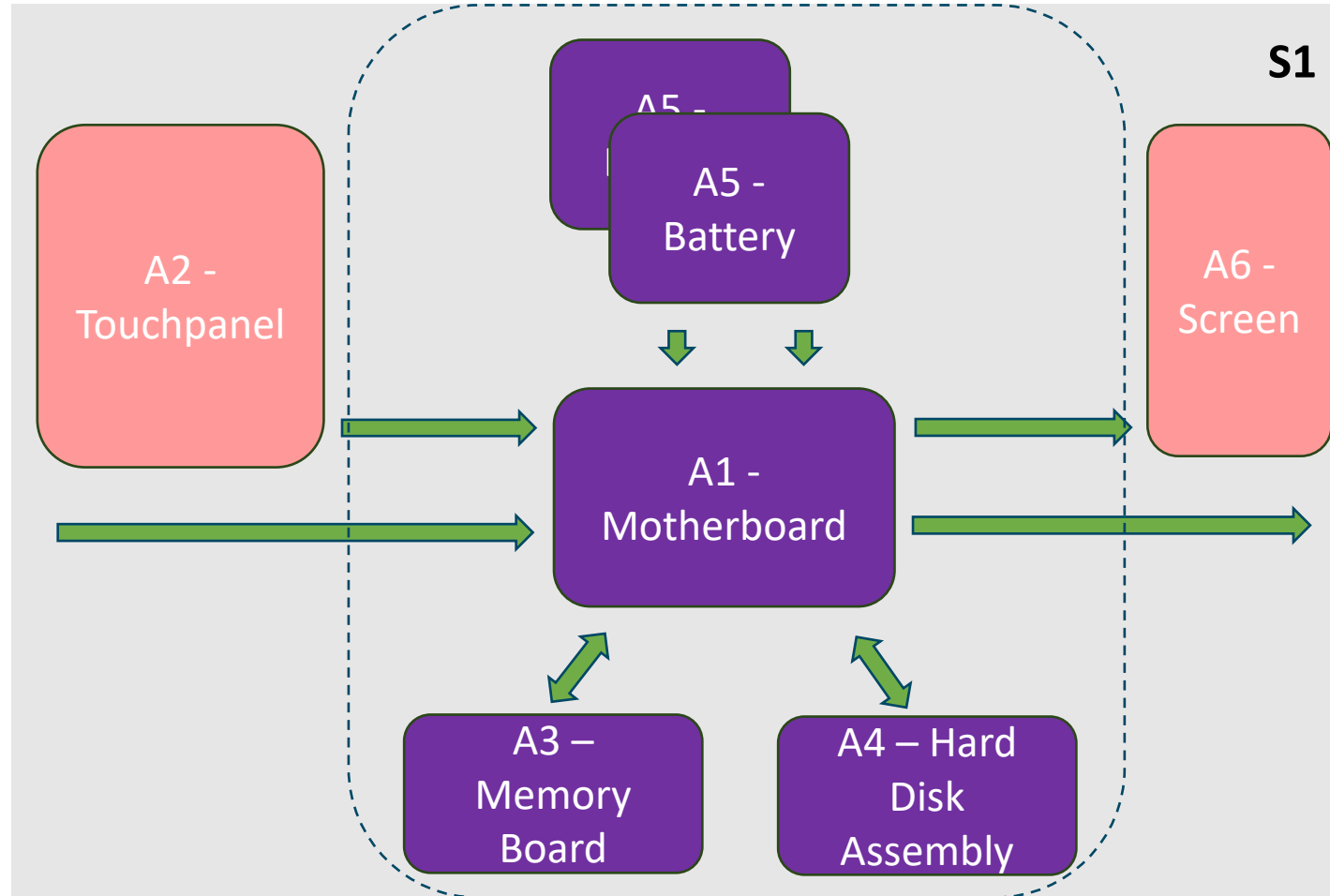
Functions and signals are determined.



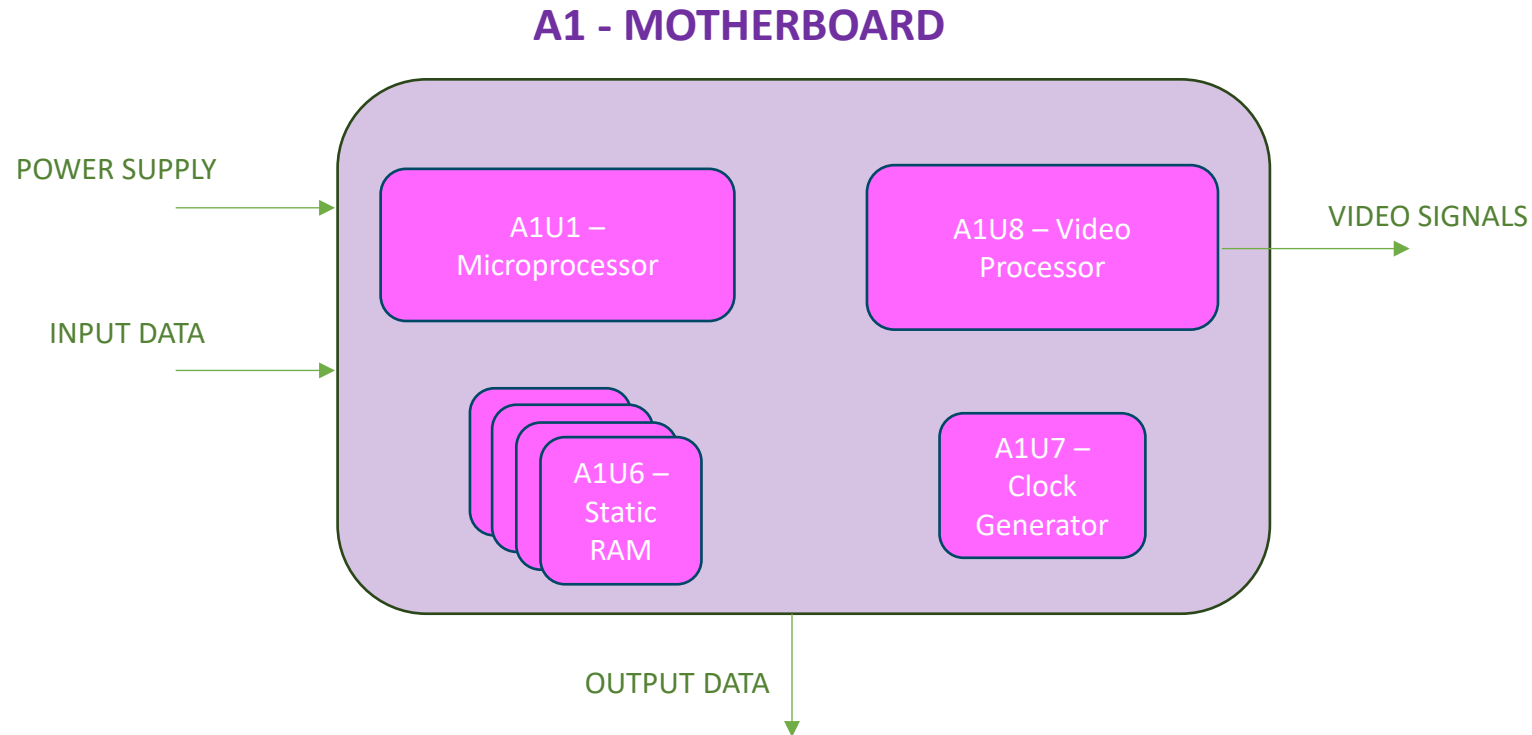
PPFMEA Examples – FFMEA



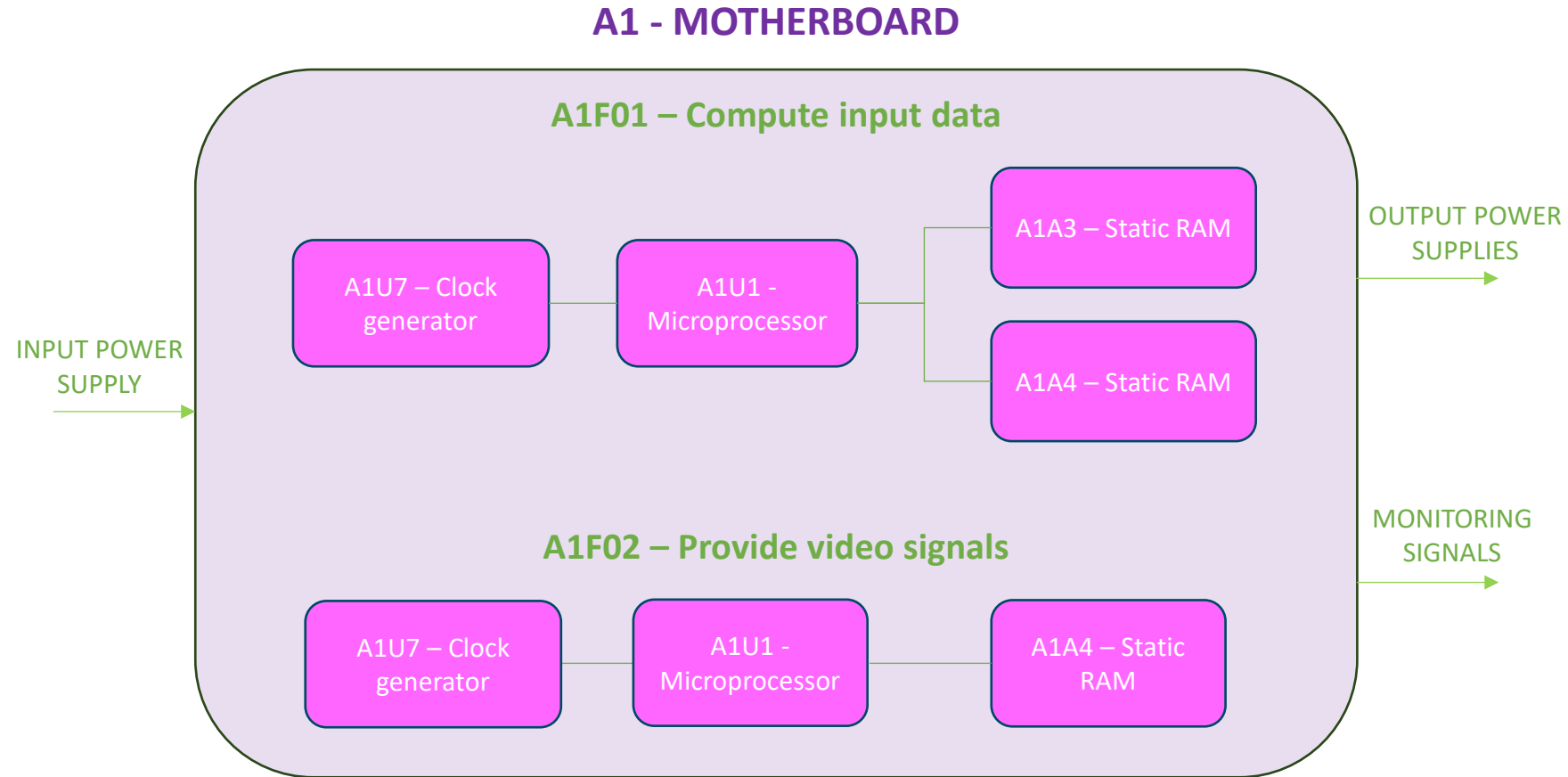
PPFMEA Examples – Indenture Level 1-2



PPFMEA Examples – Indenture Level 2-3



PPFMEA Examples – Reliability Model



PPFMEA Examples – PPFMEA Spreadsheet (1)

Item	Function	Failure Mode	Local Effect	End Effect	Sev	Item Failure Rate	Mode Percentage	Mode Failure Rate
Motherboard	Provide electrical connections for components of system to communicate	Degraded processor functionality	Loss of some or all motherboard functions	Degraded tablet pc function	7	5,24045	32.09	1,68141
Motherboard	Provide electrical connections for components of system to communicate	Loss of processor	Loss of motherboard	Tablet PC failure	9	5,24045	57.04	0,98918
Motherboard	Provide electrical connections for components of system to communicate	Data loss	Loss of motherboard	Tablet PC failure	9	5,24045	4.31	0,22567
Motherboard	Provide electrical connections for components of system to communicate	Loss of 60 ns SRAM	Loss of motherboard	Tablet PC failure	9	5,24045	5.48	0,28722
Motherboard	Provide electrical connections for components of system to communicate	No signal	Loss of clock generator	Tablet PC failure	7	5,24045	0.23	0,01231
Motherboard	Provide electrical connections for components of system to communicate	Degraded video display	Degraded video display	Inconvenience to user	7	5,24045	0.20	0,01044
Motherboard	Provide electrical connections for components of system to communicate	Loss of video display	Loss of video display	Tablet PC failure	9	5,24045	0.35	0,01856
Motherboard	Provide electrical connections for components of system to communicate	Signal stuck high or low	Loss of clock generator	Tablet PC failure	7	5,24045	0.30	0,01566
Touchscreen	Primary input device	No response to touch	Touchpanel inoperable	Tablet PC failure	9	6,55000	50.00	3,27500
Touchscreen	Primary input device	Inaccurate touch position	Loss of control on touchpanel	Inconsistent communication from touchpanel	7	6,55000	50.00	3,27500
Memory Board	Provide storage for processor results	Degraded DRAM controller functionality	Loss of some or all memory board functions	Tablet PC failure	9	1,35512	25.47	0,34510
Memory Board	Provide storage for processor results	Loss of DRAM controller functionality	Loss of memory board	Tablet PC failure	9	1,35512	45.27	0,61351
Memory Board	Provide storage for processor results	Data loss	Loss of memory board	Tablet PC failure	9	1,35512	12.87	0,17447
Memory Board	Provide storage for processor results	Loss of 31ns DRAM	Loss of memory board	Tablet PC failure	9	1,35512	16.39	0,22205
Hard Disk Assembly	Store data	Loss of RAID Controller	Loss of hard disk assembly	Probable data loss	9	7,72933	2.97	0,22933
Hard Disk Assembly	Store data	No data transfer	Loss of hard disk assembly	Probable data loss	9	7,72933	32.02	2,47500
Hard Disk Assembly	Store data	Loss of hard disk	Loss of hard disk assembly	Probable data loss	9	7,72933	65.01	5,02500
Microprocessor	Control primary functions of tablet PC	Degraded Operation	Degraded processor functionality	Degraded tablet pc function	7	4,67059	36.00	1,68141
Microprocessor	Control primary functions of tablet PC	Shorted or Opened	Loss of processor	Tablet PC failure	9	4,67059	64.00	2,98918
Static RAM	Store data	Data Bit Loss	Data loss	Tablet PC failure	9	0,51289	44.00	0,22567
Static RAM	Store data	Short or Open	Loss of 60 ns SRAM	Tablet PC failure	9	0,51289	56.00	0,28722
Clock generator	Produce timing signal	Output Stuck High or Low	Signal stuck high or low	Tablet PC failure	7	0,02797	56.00	0,01566
Clock generator	Produce timing signal	Input or Output Open	No signal	Tablet PC failure	7	0,02797	44.00	0,01231



PPFMEA Examples – WRR (1)

Windchill Risk and Reliability - [Tablet PC, System: Tablet PC]

File Edit View Insert Tools System Prediction Filter Tree Window Help

System Tree Items

Name	Part Number	System Tree Ident...	Reference Designat...	Description	Manufacturer	Failure Rate, Pred...	MTBF, Predic...	Tag
Industrial Tablet PC	PC070101	S1	S1	Tuf-Tablet	Tuf-Tablet International	25.360293	39432	
Battery	BAT56A04	A5	A5	Li-Ion tablet battery	Superior Power Devices	4.485400	222946	
Motherboard	MB060415	A1	A1	Motherboard	iTronics	5.240445	190823	
Microprocessor	MIC870A	A1U1	U1	Dual core microprocessor	AB Electronics	4.670594	214106	
Static RAM	SRAM031	A1U3-U6	U3-U6	60 ns SRAM	Memory Unlimited	0.512888	2e+06	
Clock Generator	CLK504	A1U7	U7	DDS-based clock generator	AB Electronics	0.027967	4e+07	
Video Processor	VP899011	A1U8	U8	Digital video processor	AB Electronics	0.028996	3e+07	
Touchpanel	TP55401A	A2	A2	12.1" touch panel	Clear Display Technologies	6.550000	152672	
Memory Board	MEM061789	A3	A3	DRAM memory board	Tuf-Tablet International	1.355120	737942	
DRAM Controller	DRAMC7001	A3U1	U1	DRAM controller board	Memory Unlimited	0.958604	1e+06	
Dynamic RAM	DRAM512-31	A3U2-U3	U2-U3	31ns DRAM	Memory Unlimited	0.396516	3e+06	
Hard Disk Assembly	HD061455	A4	A4	Hard disk assembly	Tuf-Tablet International	7.729327	129377	
RAID Controller	RAID-023C	A4U1	U1	SATA RAID Controller	Megastorage Corporation	0.229327	4e+06	
Hard Disk	HD70A5-500	A4HD1-HD2	HD1-HD2	500 GB Hard Disk	Megastorage Corporation	7.500000	133333	
Screen		A6				0.000000	0	

System Tree Items FMEA Table FMEA Tree Items - System Problems

FMEA Worksheet

Item / Process...	Function / Process	Failure Mode	Local Effect	End Effect	Sev	Item Failur...	Mode Pe...	Mode Failur...	Class	Cause of Failure	Occ	Current Controls
14		Degraded processor	Loss of some or all motherboard functions	Degraded tablet pc function	7		32.09	1.681414			2	



PPFMEA Examples – WRR (1)

Screen											A6		0.000000		0	
System Tree Items											FMEA Table		FMEA Tree Items - System		Problems	
FMEA Worksheet																
Filter																
	Item / Process...	Function / Process	Failure Mode	Local Effect	End Effect	Sev	Item Failur...	Mode Percen...	Mode Failure...	Class						
14	Motherboard	Provide electrical connections for components of system to communicate	Degraded processor functionality	Loss of some or all motherboard functions	Degraded tablet pc function	7	5.240445	32.09	1.681414							
15			Loss of processor	Loss of motherboard	Tablet PC failure	9		57.04	2.989180							
16			Data loss	Loss of motherboard	Tablet PC failure	9		4.31	0.225671							
17			Loss of 60 ns SRAM	Loss of motherboard	Tablet PC failure	9		5.48	0.287217							
18			No signal	Loss of clock generator	Tablet PC failure	7		0.23	0.012306							
19			Degraded video display	Degraded video display	Inconvenience to user	7		0.20	0.010438							
20			Loss of video display	Loss of video display	Tablet PC failure	9		0.35	0.018557							
21			Signal stuck high or low	Loss of clock generator	Tablet PC failure	7		0.30	0.015662							
22			No response to touch	Touchpanel inoperable	Tablet PC failure	9		50.00	3.275000							
23	Touchscreen	Primary input device	Inaccurate touch position	Loss of control on touchpanel	Inconsistent communicati... from touchpanel	7	6.550000	50.00	3.275000							
24	Memory Board	Provide storage for processor results	Degraded DRAM controller functionality	Loss of some or all memory board functions	Tablet PC failure	9	1.355120	25.47	0.345098							
25				Loss of DRAM controller functionality	Loss of memory board	Tablet PC failure		9	45.27	0.613507						
26				Data loss	Loss of memory board	Tablet PC failure		9	12.87	0.174467						
27			Loss of 31ns DRAM	Loss of memory board	Tablet PC failure	9		16.39	0.222049							
28	Hard Disk Assembly	Store data	Loss of RAID Controller	Loss of hard disk assembly	Probable data loss	9	7.729327	2.97	0.229327							
29				No data transfer	Loss of hard disk assembly	Probable data loss		9	32.02	2.475000						
30				Loss of hard disk	Loss of hard disk assembly	Probable data loss		9	65.01	5.025000						
Parts Table																
General Data																
Calculation Data																
FMEA Worksheet																
FMEA Control Plans																
FMEA DVPs																
FMEA																
Problems Form																
Ready																



References

- ✓ AIAG VDA FMEA Handbook 5th ed.
- ✓ ECSS-Q-ST-30-02C 2009 – Failure modes, effects (and criticality) analysis (FMEA/FMECA)
- ✓ GSFC-HDBK-8004 – (NASA) Guideline for Failure Modes and Effects Analysis and Risk Assessment.
- ✓ IEC 60812:2018 Failure modes and effects analysis (FMEA and FMECA).
- ✓ MIL-STD-1629 - Procedures for Performing a Failure Modes, Effects, and Criticality Analysis.
- ✓ SAE ARP5580, Recommended Failure Modes and Effects Analysis (FMEA) Practices for Non-Automobile Applications



Thank you!!!

